



USERS GUIDE

Radiant QuikBolus

DICOM RT Structure Set in, watertight 3D-printable bolus STL out.



Radiant QuikBolus

3D-printable bolus, straight from a structure set

OVERVIEW

What QuikBolus is.

Radiant QuikBolus turns a DICOM RT Structure Set into a watertight, 3D-printable bolus STL in minutes — no CAD step, no manual mesh repair, no failed prints from non-manifold geometry.

Who it's for

- Clinical physicists and dosimetrists who fabricate custom bolus in-house.
- Programs running in-house 3D printing for surface-dose treatment sites.
- Anyone who would otherwise reach for the planning-system contour export and then a CAD tool.

QuikBolus works directly from the RT Structure Set your treatment planning system already produced, reconstructs the contoured bolus volume as a solid mesh, welds a flat base plate of your chosen thickness, and exports a printable STL guaranteed to be sealed and manifold.



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QUIKBOLUS INTERFACE — DICOM RT STRUCTURE SET IN, PRINTABLE STL OUT.

Access and a first conversion.

Open the app

QuikBolus is deployed at quikbolus.radiant-mpc.com. Sign in with the access password provided in your agreement.

First conversion

1. Drag the RT Structure Set DICOM (RS*.dcm) onto the upload zone.
2. QuikBolus auto-detects the Body/External and Bolus ROIs — confirm the selection if multiple bolus contours are present.
3. Pick the base-plate thickness (typically 2-5 mm) and orientation.
4. Click *Build STL*. QuikBolus rasterizes the contours, interpolates inter-slice gaps, runs marching cubes, welds the base plate, and remeshes any non-sealed result into a manifold solid.
5. Preview the bolus against the patient body surface in the 3D viewer, then download the STL.

**QuikBolus**
RADIANT MEDICAL PHYSICS CONSULTING

3D PRINTABLE BOLUS GENERATOR · LOCAL-FIRST

✓ Upload RS

2 ROIs & Options

3 Generate & Preview

STRUCTURE SET LOADED

Confirm ROIs & set bolus options

Source: sample_RS.dcm (bundled demo) · Patient: DEMO^PATIENT. QuikBolus auto-detected the Body / External and Bolus ROIs — override them with the dropdowns if the detection is wrong.

Region of Interest

Body / External ROI

[1] BODY

The patient surface. Reconstructed for the 3D preview overlay.

Bolus ROI

[2] Bolus

The structure exported as the printable bolus solid.

Bolus Options

Voxel size (mm)

1.0

1.0 is the default. 0.5 gives a finer surface but uses ~8× more memory and time.

☒ Add base plate

Base thickness (mm)

2.0

Thickness of the flat plate welded onto the air-side face.

☒ Binary STL output

☒ Guarantee watertight

Generate Bolus STL →

Start over

SOLID MESH RECONSTRUCTION WITH BASE PLATE.

What the pipeline does, step by step.

1. RT Structure Set import

QuikBolus reads the RS DICOM and walks the ROI dictionary. The Body/External ROI is the patient surface. The Bolus ROI is the target volume to print.

2. Voxel rasterization

Each axial contour is rasterized onto a regular voxel grid; gaps between slices are filled by linear interpolation so the contour is solid, not hollow.

3. Marching cubes & smoothing

Marching cubes converts the voxel volume to a triangular mesh. Surface smoothing removes the rasterization staircase without losing volumetric accuracy.

4. Base plate fitting

A flat base plate of the requested thickness is welded onto the air-side face with a clean boolean union — the plate is what the printer prints on.

5. Watertight guarantee

The final mesh is voxel-remeshed if any holes or non-manifold edges remain, so the exported STL is always closed and printable.

6. STL export + 3D preview

Export as binary or ASCII STL. The in-browser 3D viewer shows the bolus against the body surface so you can confirm fit before sending the file to the slicer.

1. Upload ROIs 2. ROIs & Options 3. Generate & Preview

STRUCTURE 6v3, CRAS 6v3

Confirm ROIs & set bolus options

[Source: sample_R5_000 (bundle demo) - Patient: 0000000000] QuikBolus auto-detected the Body / External and Bolus ROIs — override them with the dropdowns if the detection is wrong.

Region of Interest

Body / External ROI

[1] BODY

Bolus ROI

[2] Bolus

Voxelizing contours, marching cubes, watertight remesh — this can take a minute.

Bolus Options

Voxel size (mm)

☒ Add base plate

Base thickness (mm)

2.0

☒ Binary STL output

☒ Guarantee watertight

Generating the bolus STL...

Generate Bolus STL +

The engine voxelizes the contours, runs marching cubes, and remeshes watertight

INTERACTIVE 3D PREVIEW OF THE BOLUS AGAINST THE PATIENT SURFACE.

What ships with each run.

- **Bolus STL** (binary, watertight) — drop straight to the slicer.
- **Clinical stats report** — dimensions, volume, surface area, and the build parameters used.
- **Branded 3D viewer** — a single HTML file you can send to the dosimetrist or therapist for fit review without sharing the STL.

Verification. Every generated bolus STL must be reviewed by a Qualified Medical Physicist for geometric accuracy, fit, and the water-equivalence of the print material before clinical use.

Or print a mold and cast Flexibolus.

For thick chest-wall bolus, electron treatments, or anywhere water-equivalence matters most, the same QuikBolus pipeline can design a two-part rigid mold for casting Flexibolus or comparable pourable tissue-equivalent silicone. The mold workflow lives at [/mo1d](#).

Take any watertight bolus STL (from the QuikBolus pipeline or elsewhere), upload it, and the engine generates:

- Two color-coded mold halves with a chosen parting plane — single Z, auto-suggested, or stepped to follow complex undercut anatomy.
- Automatic pour features: sprue at the cavity high point, vents at trapped-air locations, registration pegs and sockets so the halves close consistently.
- 1–3° draft on cavity walls for clean demolding, optional patient-ID embossing recessed into the cavity so casts come out labeled.
- Live undercut detection that highlights regions that won't release from the chosen parting plane — catch a bad split before you commit to a 10-hour print.
- Validation panel: wall-thickness ray-cast sampler, watertight checks, sprue position, vent count, build-volume fit against your target printer.

Export as two STLs or a single 3MF with the rigid-print profile baked in for Bambu Studio, OrcaSlicer, PrusaSlicer, or Cura. Every export also includes a Radiant-branded printable cast-info sheet covering mix ratio, pour temperature, cure window, mold release, and the validation results — ready for the cast technician.

When to choose mold over direct print. Thick bolus (> 10 mm), electron treatments where every millimetre of buildup is planned, full chest-wall coverage where direct TPU prints exceed 40 hours, or any case where water-equivalence matters more than the slight density mismatch of TPU 95A.

License, safety, and how to reach us.

License terms

QuikBolus is licensed per practice. The hosted instance at quikbolus.radiant-mpc.com is provided as a secured evaluation. The agreement covers the access password and the permitted user list.

Privacy

Uploaded RT Structure Sets are processed in-memory and on the application's working volume; nothing is retained beyond the user's session. Do not upload PHI through evaluation builds.

Contact

Live URL: <https://quikbolus.radiant-mpc.com/>

Email: info@Radiant-MPC.com

Website: radiant-mpc.com

Evaluation / decision-support only. Every Radiant tool is an aid to a Qualified Medical Physicist. All outputs must be independently reviewed before any clinical use. Do not transmit PHI through evaluation builds.